

Also, the missile's limited range, imposed by length of the wire, is not an issue in anti-tank operations. Tube-launched optically-tracked and wire-guided (TOW) of the USA with an operational range of 3750 metres and MILAN of France (Fig. 4) with an extended operational range of 3000 metres (MILAN ER) are well-known in-service wire-guided missiles.

TOW family of missiles, including TOW 2A, TOW 2B Aero and TOW Bunker Buster missiles, is the premier long-range, heavy-assault-precision anti-armour, anti-fortification and anti-amphibious-landing weapon system. TOW missiles have been integrated on ground, vehicular and helicopter platforms. MILAN is a portable medium-range anti-tank missile, with later versions equipped with tandem heat warheads, making it more effective against reactive armour.

Beam-riding weapons. Beam-riding weapons make use of a radar or laser beam for guidance. A narrow radar or laser beam is directed at the target, usually a tank or an aircraft. The missile is launched in the direction of the target and sometimes, after it is launched, it flies into the radar or laser beam. With the help of sensors and a computer onboard the missile, it keeps itself within the beam. The aiming station keeps the beam pointed at the target till it hits it. The inherent shortcoming of the radar-beam-riding guidance is that the beam spreads as it moves away from the aiming station. Laser beam riders do not have this limitation.

Beam-riding guidance is generally used for short-range air-defence and anti-tank-guided missile applications. LAHAT (short-range anti-tank missile manufactured by Israel Aerospace Industries) (Fig. 5), Starstreak (short-range air-defence system manufactured by Thales Air Defence) and RBS-70 (short-range anti-tank guided missile manufactured by SAAB Bofors Dynamics) are known laser beam riders.

LAHAT is an advanced laser-homing attack missile that makes use of semi-active laser guidance.



Fig. 4: MILAN missile



Fig. 5: LAHAT ATGM



Fig. 6: RBS-70NG laser beam rider

The target in this case can be designated either directly from the launching platform or by another land based or aerial platform. It has an operational range of eight kilometres when fired from a ground platform and 13km when fired from an aerial platform, and has a hit accuracy of 0.7m CEP. It is in use in Germany, Israel and India.

Starstreak is a man-portable/vehicle-mounted high-velocity missile with 3.5Mach velocity, designed to counter threats from conventional air threats and fast pop-up strikes by helicopter attacks. It has an operational range in excess of seven kilometres and is currently in the inventory of armed forces of the UK, South Africa, Indonesia and Thailand.

RBS-70 is not susceptible to any deception by countermeasures employed by the target aircraft in

the form of chaff or flares. It has an operational range of zero to six kilometres and a speed up to 1.6Mach. RBS-70 new-generation (RBS-70NG) includes an improved sighting system capable of night vision. RBS-70 Mk-2 upgrade is called Bolide missile. It is faster with a speed of 2Mach against 1.6Mach of standard RBS-70, and a range of eight kilometres as against six kilometres in the case of standard RBS-70. Fig. 6 shows RBS-70NG.

GPS/INS-guided weapons. GPS/INS-guided weapons make use of a multi-channel GPS receiver to provide information on the weapon's location and an inertial measurement unit (IMU) to monitor its attitude to adjust its flight path to precisely hit the target. This is a low-cost means of precision targeting that is unaffected by weather and target concealment and is immune to countermeasures. These weapons are primarily used against fixed targets or relocatable targets whose location is likely to remain static for the duration of planning and attack.

In another role, GPS/INS guidance is also used to adjust a weapon's free fall to hit a predefined point fed into the weapon, prior to launch. Yet another application of GPS guidance is in mid-course correction of guided missiles and cruise missiles. Precision in the basic weapons is characterised by a CEP in the range of one to ten metres. However, CEP is considerably improved when GPS is used together with a semi-active laser (SAL) or imaging infra-red (IIR) seeker head.

Guidance techniques

Guided weapons may use more than one guidance mechanisms for improved performance. The different guidance techniques include:

1. Beam rider guidance
2. Command guidance
3. Homing guidance
4. Navigation guidance

Homing guidance further comprises: semi-active homing guidance, active homing guidance, passive